

Practical -10

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GNU nano 8.7.1 mmap_compare.c
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <string.h>
#include <time.h>

#define SIZE 4096

double get_time()
{
    struct timespec start;
    clock_gettime(CLOCK_MONOTONIC, &start);
    return start.tv_sec + start.tv_nsec / 1e9;
}

int main()
{
    int fd;
    char *mapped;
    char buffer[SIZE];

    printf("==== Memory Mapped File I/O =====\n");

    /* Create and write initial data */
    fd = open("mmap_file.txt", O_RDWR | O_CREAT | O_TRUNC, 0666);

    if (fd < 0)
    {
        perror("open");
        return 1;
    }

    memset(buffer, 'A', SIZE);

    if (write(fd, buffer, SIZE) != SIZE)
    {
        perror("write");
        close(fd);
    }
}
```

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    return 1;
}

/* Memory Mapping */
double start = get_time();

mapped = mmap(NULL, SIZE,
              PROT_READ | PROT_WRITE,
              MAP_SHARED, fd, 0);

if (mapped == MAP_FAILED)
{
    perror("mmap");
    close(fd);
    return 1;
}

/* Modify file through memory */
strcpy(mapped, "Hello from mmap() in Ubuntu!");

/* Write changes to file */
if (msync(mapped, SIZE, MS_SYNC) == -1)
{
    perror("msync");
}

munmap(mapped, SIZE);

double end = get_time();

printf("Data written using mmap():\n");
printf("%s\n", buffer);

printf("\nMemory-mapped I/O time: %.9f seconds\n",
        end - start);

close(fd);

/* Traditional read/write */
fd = open("mmap_file.txt", O_RDONLY);
```

```
if (fd < 0)
{
    perror("open");
    return 1;
}

memset(buffer, 0, SIZE);

start = get_time();

if (read(fd, buffer, SIZE) < 0)
{
    perror("read");
    close(fd);
    return 1;
}

end = get_time();

printf("\nData read using read():\n");
printf("%s\n", buffer);

printf("\nTraditional read() time: %.9f seconds\n",
        end - start);

close(fd);

printf("\n==== Comparison =====\n");
printf("mmap(): Direct memory access to file contents.\n");
printf("read()/write(): Data is transferred using system calls.\n");

return 0;
```

Output:

[illegible]